ECON 1550: International Finance

Models in Economics

Bottom line to remember

“All models are wrong, some models are useful”

Example: Uncovered Interest Parity (UIP)

$$R_{\$} = R_{¥} + \frac{E^e_{\$/¥} - E_{\$/¥}}{E_{\$/¥}}$$

- Does not fit the data well
- We can still use it to determine $E_{\$/¥}$ given $R_{\$}$, $R_{¥}$, and $E^e_{\$/¥}$
- Combined with goods and money market equilibrium, we can understand (some or all) effects of monetary policy on $E_{\$/¥}$

Example: UIP (continued)

$$R_{\$} = R_{¥} + \frac{E^e_{\$/¥} - E_{\$/¥}}{E_{\$/¥}} + rp$$

- How much did we miss? Add an error term and call it **risk premium**
- If rp is independent of monetary policy, we did not miss anything
- If rp depends on monetary policy, we still captured one channel

Types of Variables

Endogenous	• Explained within the model
Exogenous	• Taken as given • Not explained by the model
Parameters	• Exogenous; do not depend on policy

Types of Equations

Identities	• Hold by definition or construction
Behavioral	• Capture behavior that we include in a model • Hold by assumption
Equilibrium conditions	• Supply equals demand • Hold by "economic logic"

Solving a Model, Solving for a Variable

- “Solving for a variable” means expressing that variable in terms of exogenous variables only
- “Solving a model” means solving for all endogenous variables

Example 1

Exogenous variables

Variable	Description
T	taxes
Y	income
c_1	marginal propensity to consume

Endogenous variables

Variable	Description	Equation	Type of equation
C	consumption	$C = c_1 Y_D$	behavioral
Y_D	disposable income	$Y_D \equiv Y - T$	identity

Solution and Intuition

T GOES UP

$Y_D \equiv Y - T$

$C = c_1 Y_D = c_1 (Y - T)$

① USE SOL

$T\uparrow \quad Y_D\downarrow \quad C\downarrow$

② USE ORIGINAL EQ. TO EXPLAIN

Example 2

Exogenous variables

Variable	Description
C	consumption
Y	income
c_1	marginal propensity to consume

Endogenous variables

Variable	Description	Equation	Type of equation
T	taxes	$Y_D \equiv Y - T$	identity
Y_D	disposable income	$C = c_1 Y_D$	behavioral

Solution and Intuition

Example 2

Exogenous variables

Variable	Description
C	consumption
Y	income
c_1	marginal propensity to consume

Endogenous variables

Variable	Description	Equation	Type of equation
T	taxes	$Y_D \equiv Y - T$	identity
Y_D	disposable income	$C = c_1 Y_D$	behavioral

Solution and Intuition

$C\uparrow \quad Y_D\uparrow \quad T\downarrow$

$$\boxed{Y_D = \frac{1}{c_1} C}$$

$$\frac{1}{c_1} C = Y - T \Rightarrow \boxed{T = Y - \frac{1}{c_1} C}$$

1 $C\uparrow \quad Y_D\downarrow \quad T\downarrow$

2

Shocks

- Changes in exogenous variables
- Including changes in parameters
- Usually unforeseen, unforeseeable, or random